

Digital and Service Trade

數位與服務貿易

Wen-Bin Chuang
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Digital Trade

Digital trade includes digitally ordered goods/services and especially digitally delivered services (e.g., software, cloud computing, streaming, e-commerce, data flows, ICT-enabled services). It includes cross-border data flows.

Main Categories of Digital Trade

Category	Examples	Importance
Digitally Delivered Services	Software, cloud computing, streaming, telemedicine, online education	Fastest growing
Digital Products	e-books, music, software downloads, 3D printing files	Fully digital
E-commerce	Cross-border online sales (Amazon, Shopee, Temu)	Goods + digital
Data Flows	Movement of personal, commercial, and government data	Backbone of digital economy

Key Economic Characteristics

- **Near-zero marginal cost:**
 - Once created, digital goods/services can be replicated and delivered at almost zero extra cost.
- **Network Effects** and **Platform Economies** (winner-takes-most dynamics).
- **Scalability:** Global reach with minimal physical infrastructure.

Main Economic Effects:

- **Trade Cost Reduction:** Digital delivery dramatically lowers trade costs compared to physical goods.
- **Services Trade Boom:** Traditionally non-tradable services (e.g., consulting, education, healthcare) become tradable.
- **Market Expansion for SMEs:** Small firms can export directly via platforms (e.g., Etsy, Shopify, Temu).
- **Data as a Factor of Production:** Similar to how capital and labor drive growth.

Economic Models

- **Endogenous Growth Theory:** Digital trade accelerates **idea flows** and innovation.
- **Gravity Model Extensions:** Digital trade reduces “distance” dramatically (time and cost).
- **Baumol’s Cost Disease:** Services that were non-tradable become tradable → productivity gains in services sector.

Key Issues in Digital Trade

- **Cross-border Data Flows** and localization requirements
- **Data Privacy** (GDPR in EU vs lighter rules elsewhere)
- **Cybersecurity and National Security**
- **Digital Taxation** (e.g., Digital Services Taxes)
- **Intellectual Property Protection** (piracy, counterfeiting)
- **Platform Regulation** (Big Tech dominance)

Major Agreements

- **CPTPP** and **DEPA** (Digital Economy Partnership Agreement): High-standard digital rules
- **USMCA** (Chapter 19): Strong digital trade chapter
- **EU** approach: More regulatory (data protection first)
- **WTO** negotiations on e-commerce (ongoing)

Service Trade

Services trade (e.g., finance, tourism, transport, professional services, telecom) differs significantly from goods trade. It was traditionally harder to measure and model than goods trade due to its **intangibility and regulatory barriers**.

- **Better fit than for goods:** Kimura and Lee (2006) found that gravity equations explain services trade better than goods trade. Their adjusted R^2 values were notably higher for services (around 0.80) compared to goods (around 0.65).

- **Core drivers:**
 - **Economic size (GDP):** Strongly positive, as expected – larger economies trade more services.
 - **Common language:** One of the most robust and important positive factors (often stronger than for goods, due to the personal/cultural nature of many services).
 - **Geographic distance:** Frequently **insignificant or weaker** than in goods trade. This makes intuitive sense because many services (especially Mode 1 – cross-border supply via digital means) do not require physical transportation.
- **Other factors:** Cultural proximity, colonial ties, common legal systems, and trade agreements boost services flows. Regulatory barriers have a strong negative impact.

Basic Gravity Model (Traditional Goods Trade)

The **classic** gravity equation is:

$$T_{ij} = G \frac{Y_i^\alpha Y_j^\beta}{D_{ij}^\theta}$$

Where:

- T_{ij} = Trade flow from country i to country j
- Y_i, Y_j = Economic size (GDP) of country i and j
- D_{ij} = Distance between i and j , G = Constant
- α, β, θ = Parameters to be estimated

In **log-linear form** (most commonly used for estimation):

$$\ln(T_{ij}) = \alpha + \beta_1 \ln(Y_i) + \beta_2 \ln(Y_j) + \beta_3 \ln(D_{ij}) + \epsilon_{ij}$$

Key Modified Gravity Equation for Services:

Services trade follows the gravity model **quite well**, but with some important differences compared to goods trade.

$$\ln(S_{ij}) = \alpha + \beta_1 \ln(Y_i) + \beta_2 \ln(Y_j) + \beta_3 \ln(D_{ij}) \\ + \beta_4 \text{Cultural}_{ij} + \beta_5 \text{Regulatory Similarity}_{ij} + \beta_6 \text{Mode Dummies} + \epsilon$$

Empirical Findings:

- Distance elasticity (β_3) is typically **lower** for services than for goods (services are less sensitive to physical distance).
- **Cultural proximity** and **institutional quality** matter more for services.

Important Variables Added for Services:

Variable	Effect on Services Trade	Reason
Distance (D_{ij})	Negative (but weaker than goods)	Many services are delivered digitally or via Mode 3 (FDI)
GDP / Market Size	Strongly Positive	Larger economies trade more services
Common Language / Colonial Ties	Strongly Positive	Very important for services
Regulatory Similarity	Positive	Similar regulations reduce barriers
Digital Infrastructure	Positive	Broadband, 5G, data centers
Trade Restrictiveness (STRI)	Negative	Higher restrictions → lower trade
Mode of Supply Dummies	Varies	Mode 1 & 3 usually larger

Differences from Goods Trade

Aspect	Goods Trade	Services Trade
Distance effect	Strong negative	Weaker or insignificant
Language/Culture	Important	Often more critical
Main barriers	Tariffs, transport costs, borders	Regulations, licensing, data rules, FDI restrictions
Data quality	Better measured	Patchy, missing values, harder to observe
Model fit	Good	Often better
Modes of supply	Mainly physical	GATS Modes 1-4 (cross-border, consumption abroad, commercial presence, movement of persons)

- Services trade involves more "dark costs"
 - opaque regulations in sectors like professional services, finance, and telecom – compared to the more transparent tariffs on goods.

Key findings from studies:

- **Economic size (GDPs)** remains the strongest positive driver.
- **Distance** has a significant negative effect (though sometimes weaker than for goods, due to lower physical transport needs).
- **Common language**, **cultural proximity**, and **institutional similarities** are particularly important.
- Services trade restrictions (measured by indices like the OECD **STRI**) strongly reduce flows.
- The model successfully explains why services trade is highly concentrated among high-income countries and is sensitive to regulatory barriers.

Services Trade Restrictiveness Index(STRI)

- The **STRI (經合組織服務貿易限制指數)**, developed by the OECD, is a groundbreaking tool because it quantifies what was historically unquantifiable: **non-tariff regulatory barriers in services**. Unlike goods, where tariffs are the primary barrier, services trade is restricted by domestic regulations (e.g., licensing, foreign equity limits, visa quotas)..
- To understand its impact on trade, we must first understand what STRI measures. It scores countries from **0 (completely open)** to **1 (completely closed)** across 22 service sectors based on five policy dimensions:
 1. **Foreign Ownership Restrictions:** Limits on foreign equity, screening mechanisms, or bans on foreign land ownership.
 2. **Movement of People:** Visa quotas, economic needs tests, and lack of mutual recognition of professional qualifications.

3. **Other Discriminatory Measures:** Subsidies only for domestic firms, public procurement restrictions, or forced technology transfer.
4. **Barriers to Competition:** State-owned monopolies, exclusive licenses, or limits on the number of operating firms.
5. **Regulatory Transparency:** Lack of public consultation, unclear licensing procedures, or lack of independent regulatory bodies.

Integration with the Gravity Model

- Because services don't face **border tariffs**, economists use STRI to calculate the **Ad Valorem Equivalent (AVE)**
 - Essentially translating a regulatory restriction into a "tariff equivalent."
 - *Example:* An OECD study found that in some highly restricted sectors (like certain **telecommunications** or **professional services**), the STRI score translates to an AVE of **30% to over 100%**. This means the domestic regulation acts exactly like a 100% tariff, severely choking trade.

When applied to the Gravity Model, the equation looks like this:

$$\ln(Trade_{ij}) = \beta_0 + \beta_1 \ln(GDP_i) + \beta_2 \ln(GDP_j) \\ + \beta_3 \ln(Distance_{ij}) - \beta_4(STRI_i + STRI_j) + \epsilon$$

Empirical Findings:

- **Elasticity:** A 10% reduction in a country's STRI score is empirically shown to increase its services imports by roughly **3% to 5%**, depending on the sector.
- **Symmetry:** The STRI of **both** the exporting and importing country matters.
 - If the exporter has high STRI (e.g., poor regulatory transparency), it raises costs for their firms, reducing their export competitiveness.
- Perhaps the most critical insight regarding STRI is that services restrictions do not just hurt services trade; they hurt manufacturing and the broader economy.

- Interestingly, **Digital trade** often shows strong gravity patterns but with some nuances:
 - **Distance** matters less for purely digital delivery, while **regulatory alignment** (data privacy, cybersecurity, digital standards) and **digital readiness** become critical "distance-like" frictions.
 - That is to say, **Digital trade** (especially digitally delivered services) further weakens the role of physical distance.
- It adapts the classic framework to analyze trade in digitally ordered goods and especially digitally deliverable services (DDS), such as software, cloud computing, financial services, professional services, streaming content, telecom, and data-related flows.

The core structure remains the same:

- **Positive drivers:**

- Economic sizes (GDPs), digital readiness/connectivity.

- **Negative drivers:**

- Various forms of "distance" — geographic, regulatory, data localization rules, etc.

Differences from Traditional Services and Goods Trade

Aspect	Goods Trade	Traditional Services	Digital Trade (DDS)
Distance	Strong negative	Weaker	Weakest (but regulatory distance strong)
Key enablers	Infrastructure, tariffs	Regulations, proximity	Digital connectivity, data rules
Policy focus	Tariffs, NTBs	STRI, licensing	E-commerce chapters, data flows
Complementarities	With services	With goods	Strong with ICT goods & other services
Data challenges	Good	Patchy	Emerging but improving

- Digital trade shows **strong complementarities**: Better digital infrastructure and ICT goods imports amplify services exports.

Extended Gravity Model for Digital Trade:

$$\ln(DT_{ij,t}) = \alpha + \beta_1 \ln(GDP_{i,t}) + \beta_2 \ln(GDP_{j,t}) + \beta_3 \ln(Dist_{ij}) \\ + \mathbf{X}'_{ij,t}\boldsymbol{\gamma} + \delta_{ij} + \lambda_t + \epsilon_{ij,t}$$

Where:

- $DT_{ij,t}$ = Digital trade flow from country i to j in year t (e.g., digitally delivered services, e-commerce, etc.)
- $GDP_{i,t}, GDP_{j,t}$ = Economic size (GDP)
- $Dist_{ij}$ = Geographical distance between countries i and j
- $\mathbf{X}_{ij,t}$ = Vector of extended variables (most important part)
- δ_{ij} = Country-pair fixed effects, λ_t = Year fixed effects, $\epsilon_{ij,t}$ = Error term

Key Differences from Traditional Gravity Model:

Factor	Goods Trade	Services Trade	Digital Trade
Distance Effect	Strong negative	Moderate negative	Weak / Sometimes insignificant
Border Effect	Strong	Moderate	Very weak
Digital Connectivity	Minor	Important	Very Strong
Data Policy	Not relevant	Moderately important	Critical
Language & Culture	Important	Very Important	Still important
Regulatory Harmonization	Important	Very Important	Extremely Important

The Digital Extension: DSTRI (Digital STRI)

As trade digitized, the OECD recognized that traditional STRI didn't fully capture **digital barriers**. They introduced the **Digital Services Trade Restrictiveness Index (DSTRI)**.

New Digital Barriers Measured:

1. **Data Localization:** Requirements that data generated domestically must be stored on domestic servers (massively increases costs for cloud and digital services).
2. **Source Code Disclosure:** Forcing foreign software companies to hand over proprietary source code as a condition of market access.
3. **Digital Signatures & E-contracts:** Lack of legal recognition for cross-border electronic contracts.

Impact on Digital Trade:

Using the Gravity Model with DSTRI, researchers found that data localization requirements alone can reduce digital trade flows by **up to 10-15%**. DSTRI is now crucial for understanding the trade of "digitally deliverable services" (e.g., software, streaming, cloud computing).